

Mission Control Interview – Operational Protocols & Communication (Flight Director)

Overview

This document captures firsthand insights into mission control operations, communication protocols, and decision-making practices for flight Directors. This data has been compiled from interview data from **two** real flight directors.

Capcom

Role Description-

Capsule Communicators are astronauts assigned to communicate directly with astronauts in space during a mission. They give updates on procedures, notify timeline changes, coordinate spacewalks, and give emergency instructions when needed.

Decision Making-

Capcoms follow procedures and AREN'T the ones making the final decisions, the director has all the saying power (even above the president). They simply communicate to the crew. However, how that information gets delivered under pressure is up to them; especially when providing clarification. Capcoms also must consider the crew's perspective before relaying information to prevent panic, confusion, or stress.

Communication -

Capcoms communicate in a very structured manner, speaking directly to the astronauts. Its very direction based with serious vocal tones to communicate the gravity of the situation. Emergency procedures are rehearsed and the information must be relayed in a way that conveys trust.

Decision Making

Focus Amid Multiple Conversations

- You must focus on certain conversations that are happening when making decisions.
- It is a learned skill to be able to focus on one conversation at a time when everyone is talking to you.
- Mission Control is full of simultaneous inputs, but effective decision making requires deliberate mental filtering and disciplined attention.

Always Listen to the Crew

- You must always listen to your crew as a Flight Director.
- Crew input is critical and must never be ignored, especially when multiple communication loops are active.

Do Not Give Up Options Too Soon

- The most important thing when making decisions is to weigh your options.
- Do not give up on any options too soon.
- Preserving flexibility is essential. Closing off options prematurely can limit the ability to respond effectively as situations evolve.

Conflict Resolution

Core Principles

- Conflict in Mission Control is constant.
- It is not democratic.
- The Flight Director is the final decision-maker.
- Every Flight Controller is responsible for ensuring the decision is data-driven.
- If two controllers disagree, then you must Refine options, find compromise, and present a joint position to the Flight Director.

- Controllers must stop the Flight Director if critical data is missing, fully inform them, then implement the decision, even if they disagree.

Chain of Command and International Tension

- There are documented flight rules outlining authority.
- If international partners disagree: negotiate, minimize risk, and maintain crew safety.
- Example: An autonomous vehicle approached the station. An antenna was supposed to fold to prevent contact. It failed to fold. Operators expected a backup maneuver. The partner side refused due to their internal directives. A spacewalk was required to manually remove interference.
- You may not control others' decisions, but you control how you mitigate risk.

Training

Training Environment vs. Real Operations

- When training others as a flight director, you do not have a real-time urgency, which is different from real operations.
- Some methodologies are different from other languages when you are training others as a flight director.
- It might be difficult for others to learn a technical system that is not in their native language as a flight controller.
- Some individuals might lack a common language as well as a technical language as a flight controller. You must be aware of that when training.

Teaching Theory, Integration, and Standardization

- When training as a flight director, you must learn the theory and understand how to integrate the context and standardize it.

- As a flight director, you must teach the theory and understand how to integrate the context and standardize.
- As a flight director, you must be aware that people in training and Ops have different diagrams to convey information.

Operating in a Life-and-Death Environment

- When training as a flight director, you must learn how to properly teach and think in an environment where things are a matter of life and death. You must throw away everything you know.
- Mental toughness is important when training others as a flight director.

Priorities, Facts, and Professional Discipline

- When training others as a flight director, some priorities that you have are the safety of your crew, the integrity of the spacecraft, and to execute the mission.
- Facts are extremely important when acting on decisions. There is no hesitation or buckling under pressure.
- You must exercise confidence, methodology, and sound reasoning when training as a flight director.

Communication

Language, Interpretation, and Cross-Cultural Communication

- When communicating as a flight director, you have a big reliance on an interpreter since people may not understand your language.
- Sometimes when communicating, not everyone speaks your language. This means that you must use very specific vocabulary that gets utilized in the control center.
- When you have better relations with other communities as a flight director, you have overall better interactions with others.

Tone, Demeanor, and Authority

- As a flight director, there are different vocal tones that show how serious you are. Other people's demeanor is very important in controlling communication.
- There are different vocal tones to show how serious you are when you are a flight director. It is important in controlling the situation.

Clarity, Accuracy, and Trust

- You must always provide accurate information in a way that others can understand and trust you as a flight director.
- There is a chance for misunderstandings to happen where things are not followed directly when communicating as a flight director.

Training, Procedures, and Mission Focus

- You will be rehearsing emergency situations through training as a flight director.
- You must be aware of and follow multiple procedures as a flight controller.
- There is a need to always focus on the mission as a flight director.
- One of the biggest issues is teaching people how to communicate as a flight director in the MCC.
- Communication is a very important thing as a flight controller. It is the key to everything.

Visibility In Mission Control Room

Viewable Items In Mission Control

This a list of items that are visible to flight directors when that they utilize during operations:

- Foundation of Flight Operations (posted and even under desk glass)
- Mission patches (including Apollo-era missions)
- Multiple clocks (ISS operates on Greenwich Mean Time)

- Caution and warning displays
- Timelines
- Communication loops

End of list

- In chaos, certain elements must always be visible:
- If an action cannot be traced to the Foundation of Flight Operations, controllers must ask how it aligns.
- The Flight Director often watches the room, observes time execution, and monitors team behavior more than individual data streams.

Lived Experience

The Legacy of Apollo 1 and Operational Foundations

- After the Apollo 1 fire, flight controllers wrote **"Tough and Competent"** at the top of their chalkboards.
- The fire occurred in a 100% oxygen environment with exposed power cables.
- With people moving around in that configuration, ignition became possible.
- Everyone knew the configuration was risky, but no one forcefully stopped it.
- The result was loss of life.
- From that moment forward:
- **Competence** meant knowing your system in total depth.
- **Toughness** meant standing up in Mission Control and saying: *"This is wrong. We should not be doing this."*
- Young flight controllers were expected to know their systems better than anyone, speak up, even if it meant challenging leadership, and accept that responsibility is personal.
- The Foundation of Flight Operations became a living document, constantly referenced, constantly applied.

High-Stakes Decisions

- Flight Directors must make decisions that are fast, high-stakes, and sometimes frightening
- When “everything hits the fan,” success depends on competence, toughness, and strong teamwork
- The goal is to go home knowing you did everything possible.
- Throughout Mission Control are reminders of the cost of spaceflight such as mission patches and memorials that acknowledge the 17 lives lost over the course of human spaceflight. Everyone understands the gravity of the job.

Safety Hierarchy and Risk Philosophy

- The core concept drilled into operators are the Safety of the crew, the Integrity of the vehicle and then the Success of the mission
- If necessary, sacrifice the mission to save the vehicle and sacrifice the vehicle and mission to save the crew.
- Space is inherently risky. The job is not to eliminate risk, but to Understand it, Control it, and Decide whether the outcome justifies taking it
- Sometimes the riskier path is chosen, if it is controllable and more beneficial.

Designing for Risk

- Examples of operational risk debates such as engineers resisted placing windows near hatches because it was difficult.
- Operators wanted a direct visual (“mark one eyeball”) to verify safety.
- A camera was questioned: Is it live? Is it frozen? How do we buy down that risk?
- On ISS, you can return quickly if something fails.
- In lunar halo orbit, the worst-case return to Earth must be engineered.
- Redundancy stacks quickly—mass becomes a constraint.
- Launching in one vehicle and landing in another adds complexity.

- EVA operations differ significantly from tethered ISS EVAs.
- Rovers require communications architecture (antennas, mapping systems).

Moon vs. Mars – Shifting Control Authority

- On the ISS: Ground executes in real time and the crew performs science with constant support.
- For lunar missions, communication delay increases, crew autonomy increases, and ground becomes advisory, not executional.
- For Mars (22-minute delay) the real-time Mission Control becomes impossible.
- One concept: “Take Mission Control with you.”
- AI-driven flight control could detect anomalies, isolate the relevant data subset, prevent information drift, and reduce redundant hardware and training mass.
- The idea is not replacing humans, but embedding decision capability locally.